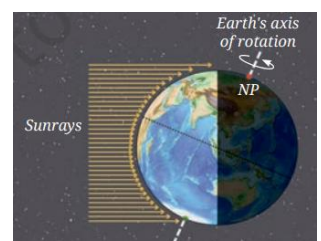
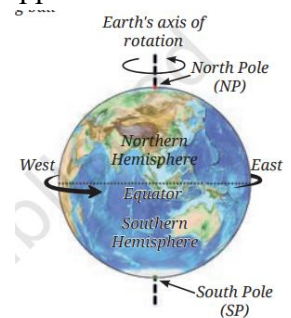


Chapter – 12: Earth, Moon, and the Sun

- 1 morning – Kanyakumari, Tamil Nadu – 12-yr old Rashmika – cycling to school
- Very excited – that day – science teacher – explain – interesting obs.
- Rashmika – notice – coconut tree shadows – long – morning – shorter – afternoon
- She thought – decided – shadow size – change – Sun moves
- BUT – remembers – learning – grade 6 – *curiosity* – chapter – Beyond Earth – Earth moves around Sun
- Now wonders – Sun, Earth – who moves?

Rotation of Earth

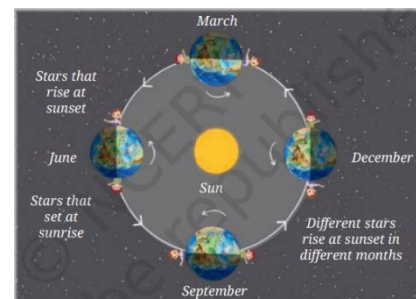
- You notice – Sun rises – East – sets – West – wonder why?
- Ever enjoy – ride – merry-go-round
- Activity –
 - Sit – merry-go-round – face – outer side
 - Ask someone – turn ride – slowly – anti-clockwise dir. – obs. – obj. around you – appear to move – which dir.?
 - Turn – anti-clockwise dir. – obj. – appear to move – opp. dir. – clockwise
 - Fix gaze – particular tree (building) – sitting – merry-go-round – anti-clockwise
- Tree – appear to move – opp. dir. – clockwise
- Tree appears – left side – moves out of view – right side
- View from Earth – Sun – appears in East – move across sky – East to West – disappears in West
- Indication – Sun moves? – OR – Earth – turning around
- Fact – Sun – appears to move – view from Earth – turning around
- Which way – Earth – turning around?
- Visualize this – recall – some obj. – turn themselves
- Similar manner – Earth – spins (**rotates**) – own axis
- **Earth's axis of rotation** – pass through – geographic North Pole, South Pole
- Earth – **1 rotation** – 24 hrs.
- Viewed from top – North Pole – Earth – rotating – anti-clockwise dir. – West to East
- Understand this – help – globe
- Used globe – grade 6 – represent Earth – identified North Pole, South Pole, Equator
- Also learnt – axis – pass through – North, South poles – grade 6 – textbook – *exploring society: India and beyond*
- Activity –
 - Use globe – represent Earth – place sticker – mark location
 - View – above North Pole – rotate slowly – anti-clockwise dir.
 - **Observe** – location – turns around – finally – comes back – original position – 1 rotation
 - Use torch – represent Sun – go to dark room – further activity
 - Shine light – torch – some dist. – 1.5 m – over globe – **notice** – half globe – receive light – other half – dark
- Half – receive light – day time – other half – night time
- India – 1st sunrise – eastern part – then – other parts
- Look – eastern part – India – rotate globe – 1 dir. – then – opp. dir.
- India – eastern part – 1st light fall – globe rotation – West to East – w.r.t. – North-South axis



- Rotate globe – West to East – obs. location – go through – cycle – day / night
- Sunrise – location – moves into light – sunset – location – moves into darkness
- Earth's rotation – West to East – day / night cycle – side – facing Sun – daytime – other side – nighttime
- **Imagine** – standing – Equator – Earth – watch sky – rotation of Earth
- Rotation of Earth – Sun – appears – rising – eastward dir. – move across sky – East to West – set – westward dir.
- Night time – stars – visible in sky
- Activity –
 - Early evening – March-May – **identify** – Big Dipper (Saptarishi), Pole Star (Dhruva Tara) – if visible – chapter – Beyond Earth – grade 6 – *curiosity*
 - Note – location, date – night sky obs. – this act. – same night
 - **Draw** – orientation – Big Dipper – w.r.t. – Pole Star (fixed tree, building – pole star – not visible) – mark time – obs. – with sketch
 - After 2 hrs. – obs. – Big Dipper – draw orientation – note down time
 - Repeat – above step – every 2 hrs. – obs. – Big Dipper – move around Pole Star
- Earth's axis – rotation – points – close to Pole Star – Northern Hemisphere
- Pole Star – appears stationery – all stars – appear – move around
- Similar to Sun – Moon – appear – rise – eastward – set – westward
- Children discuss –
 - 1st – stars, constellations – East dir. – sunset – change – diff. months
 - 2nd – this reason – told to look – certain stars, constellations – certain times of year – grade 6

Revolution of Earth

- Earth – rotates – own axis – BUT also – revolve – around Sun – learnt – grade 6
- This movement – diff. from rotation – **revolution** – motion of obj. – around another obj.
- Path – obj. use – revolve around – **orbit**
- View from top – orbit – nearly circular
- Grade 6 – *curiosity* – chapter – Beyond Earth – Solar System – elongated orbit
- Earth – **1 revolution** – 365 days, 6 hrs

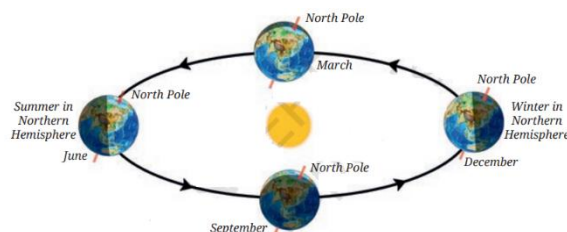


Changing view of night sky from Earth

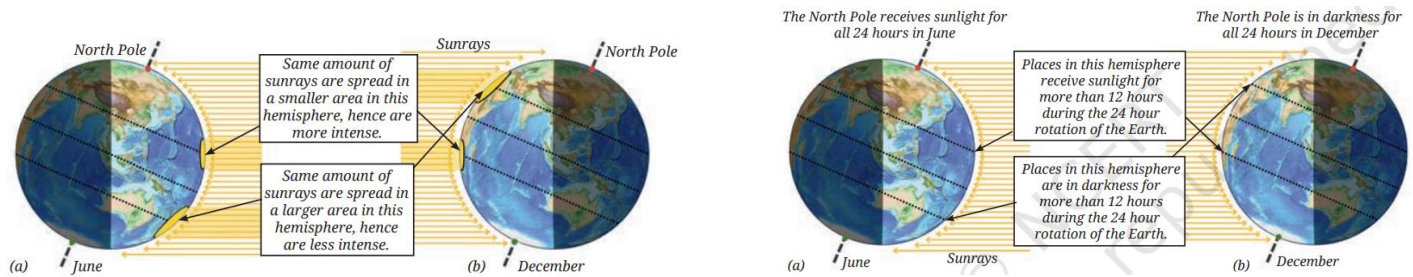
- Every evening – Sun – sets – westward dir. – night sky – visible – reason – Earth's rotation
- Earth – also revolves – around Sun – stars – visible – night sky – change gradually – vision – diff. dir.
- Also notice – this change – look – pattern of stars – fixed time – night – every month

Seasons on Earth

- Earth's axis – rotation – not upright – w.r.t. – orbit – BUT – tilted
- Earth – maintain tilt – revolution
- This tilt – add to spherical shape – give rise – seasons
- June – Northern Hemisphere – tilted – toward Sun – Southern Hemisphere – tilted – away from Sun



- Given amt. – sunrays – spread – smaller area – Northern Hemisphere – compare to Southern Hemisphere
- Northern Hemisphere – heats up – more – receive sunlight – longer duration – result – summer season
- December – opp. situation – Northern Hemisphere – winter season – sunlight – shorter time



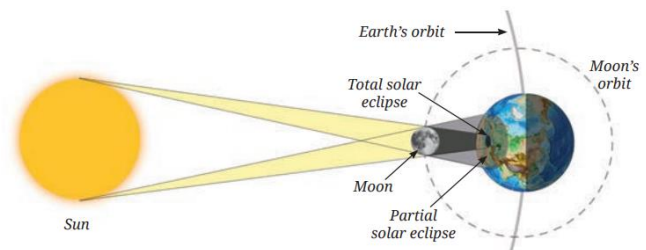
- Seasons, daytime length – reversed – Southern Hemisphere – compare to Northern Hemisphere
- Southern Hemisphere – June – winter – December – summer
- Equator – always – 12 hrs – sunlight – 12 hrs – darkness
- Little difference – intensity – sunlight – diff. months
- India – southern states – close to equator – effect – seasons – very less
- Other effects – local geographical features, proximity – oceans, seas – also influence – broader patterns – 2 hemispheres

Eclipses

- Planets – Mercury, Venus – appear – very small – compare to Sun – never block – entire light – Sun – reaching us
- BUT – moon – capable – blocking light
- Remember – grade 6 – Moon – natural satellite – Earth – revolves around Earth

Solar eclipse

- Certain times – Moon – come b/w – Sun, Earth – obstruct light – reaching Earth – **solar eclipse**
- Wonder – Moon – smaller than Sun – block sunlight
- Activity –
 - Ask friend – stand – front of you – dist. – 5 m – consider head – Sun
 - Close – 1 eye – show – thumbs up – outstretched hand – towards friend



- Can you cover – entire head – your thumb?
- Yes – entire head – covered – your thumb – thumb – smaller than head
- Size – any obj. – seen with eye – **apparent size** – depends – both – actual size, dist. from eye
- Thumb – much closer – compare to head – apparent size – both – similar
- Apparent sizes – Moon, Sun – similar – viewed from Earth
- Moon – much smaller – physical size – BUT – much closer – compare to Sun
- Moon – appear – cover entire Sun – viewed from Earth
- Fig. – arrangement – Sun, Moon, Earth – solar eclipse
- Shadow – Moon – smaller area – Earth surface
- This area – total darkness – no part – Sun – observable
- Observers – this area – witness – **total solar eclipse**
- Areas – Moon – block – partial regions – **partial solar eclipse**

- Total eclipse – few min. – turns dark – day – no sunlight – reach Earth
- Earth's rotation, motion of Moon – Moon's shadow – move across Earth's surface
- Total eclipse – visible – few min. – Moon – move away – front of Sun – partial eclipse – daylight returns
- **Safe viewing of solar eclipse**
 - **Caution** – solar eclipse – tempted – obs. – Sun – think – not intense – cause harm – BUT – even – solar eclipse – Sun – intense enough – damage eyes
 - Astronomy org. – planetaria, astronomy clubs – eclipse viewing events – solar eclipse
 - **Participate** – such events – best way – obs. – solar eclipse – organisers – provide eye protection – BUT also – scientific explanation
 - People – obs. eclipse – maintain records – ancient times
 - Times – reasons – unknown – people – feared eclipses – something – blocking main src. – light, heat – extremely concerning
 - Many superstitions – attached – solar eclipse – various parts – world – certain act. – prohibited – eating, cooking, going out
 - Newer times – reasons – known – solar eclipse – no fear – do not look – Sun – directly
 - Scientists – travel around world – obs. eclipses
 - Eclipses – provide opportunity – study phenomenon – cannot be obs. – otherwise

Lunar eclipse

- Moon – revolves around Earth – Earth – blocks Sun – reach Moon – **lunar eclipse**
- Such days – Earth's shadow – fall – Moon
- Fig. – arrangement – Sun, Moon, Earth – lunar eclipse
- Moon – completely – Earth's shadow – **total lunar eclipse**
- Bright disc – Moon – appear – dark red – stays like that – Moon – moves out
- Part of Moon – Earth's shadow – rest Moon – visible – **partial lunar eclipse**
- Unlike Sun – safely watch – eclipsed Moon – naked eye

